ADITYA UNIVERSITY

Department of Computer Science and Engineering
Academic Year (2026-27)
PROJECT TITLE REVIEW

Date of the Review	17th September 2026
Domain of the Project Work	Machine Learning
Title of the Base Paper (Including Journal Name, Volume, Issue, DoI)	**Journal Name: IEEE Access** **Year:** 2026 **Volume:** 14 **Issue:** Not assigned / N/A **DOI:** 10.1109/ACCESS.2026.3709651
Problem Statement	Students often struggle to understand their resume quality, skill gaps, job suitability, and placement chances. This project uses AI/ML to analyze resumes and student profiles. It identifies missing skills, suggests resume improvements, and matches students with relevant jobs. It also predicts placement outcomes using CGPA, skills, internships, projects, communication, and aptitude scores. The system provides personalized recommendations to improve students' career opportunities.
Proposed Title of the Project	CareerFit: AI-Based Career and Job Matching
Abstract (150–250 words)	Students often face difficulties in evaluating their resumes, identifying skill gaps, finding suitable job roles, and understanding their placement prospects. This project proposes a **machine learning-based smart career system** that combines resume analysis, skill-gap identification, job matching, and placement prediction. The system uses Natural Language Processing (NLP) to extract skills and keywords from resumes and job descriptions and compares them to identify missing or weak skills. Student factors such as CGPA, technical skills, communication skills, internships, projects, and aptitude scores are used to predict placement outcomes. Based on the analysis, the system suggests relevant skills, resume improvements, and suitable job opportunities. The proposed system aims to provide students with a personalized career profile and data-driven recommendations. By combining resume optimization, job matching, and placement prediction, the project helps students understand their current capabilities, identify areas for improvement, and prepare for suitable career opportunities more effectively.

<table>
<tr><td>Objectives and Outcomes (Two Each)</td><td colspan="3">Objectives
1. To analyze students' resumes and profiles to identify skill gaps and match them with suitable job roles.
2. To predict placement outcomes using factors such as CGPA, skills, internships, projects, communication, and aptitude scores.
Outcomes
1. A personalized analysis of the student's career profile with resume improvement and relevant job recommendations.
2. An ML-based placement prediction that helps students identify areas for improvement and prepare for suitable opportunities.</td></tr>
<tr><td rowspan="5">Details of the Students</td><td>S.No</td><td>Regd.No</td><td>Name</td></tr>
<tr><td>1.</td><td>24A95A0501</td><td>N.Lakshmi Susmitha</td></tr>
<tr><td>2.</td><td>23A91A05I3</td><td>R.L.Divya Sri</td></tr>
<tr><td>3.</td><td>23A91A05D4</td><td>V.S.S.Raghuram</td></tr>
<tr><td>4.</td><td>23A91A05J7</td><td>Swapnanil Gayen</td></tr>
</table>

Name and Signature of the Guide
with Date